

RBSE 2020 Board

Garde12

Biology

Time: 3 ¼ Hours

Max. Makes: 56

GENERAL INSTRUCTION TO THE EXAMINEES:

- (1) Candidate must write first his/her Roll No. on the question paper compulsorily
- (2) All the question are compulsory.
- (3) Write the answer to each question in the given answer-book only.
- (4) For questions having more than one part, the answers to those parts are to be written together in continuity.
- (5) If there is any error / difference / contradiction in Hindi & English versions of the question paper, the question of Hindi version should be treated valid.

(6) Section	Q. Nos.	Marks per question
A	1-13	1
B	14-24	2
C	25-27	3
D	28-30	4

- (7) Question Nos. 14,27,28,29 and 30 have internal choices.

SECTION – A

1. Write the name of four flower parts. $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = 1$
2. Marsh spot disease of pea is caused by deficiency of which element in soil ? 1
3. Write two examples of impermeable membrane. 1
4. Write the name of a growth inhibiting hormone. 1
5. Write the name of Boll worm resistant genetically modified variety of cotton. 1
6. Which part of coconut fruit is used for yielding coconut oil ? 1
7. Which vitamin is synthesised by human skin ? 1
8. Which eye pigment helps animal to see in dim light ? 1
9. Which hormone is synthesised by "C" cells of thyroid gland ? 1
10. For prevention of which disease, a person should be vaccinated with T.A.B. ? 1
11. Write an example of hinge joint. 1
12. Explain "Gastrulation". 1
13. Write the full form of E.E.G. 1

SECTION-B

14. Define bacteriophage and write one example of it. **1+1=2**

OR

Write two importance of biotechnology in medical field.

15. Write difference between estrus cycle and menstrual cycle.

1+1=2

16. Write name of a species of honey-bee and write name of two appliances of apiculture.

1 + ½ + ½ = 2

17. Explain restriction fragment length polymorphism and write one use of it.

1+ 1 = 2

18. What is biopesticide ? How Baccilus thuringiensis is used as biopesticide

1 + 1 = 2

19. Part — A contains the plant name and Part — B contains useful plant part related to them. Match the following :

Part - A

Part - B

(A) Munj

(i) Dry flower buds

(B) Clove

(ii) Leaves for fibres

(C) Black pepper

(iii) Mature Crimocarp fruit

(D) Fennel

(iv) Immature dry drupe.fruit y,

½ +½ +½ + ½ = 2

20. Explain two disorders related to heart and circulatory system. **1+1=2**

21. Explain nomenclature of enzyme on the basis of chemical reaction. **2**

22. What is Kliekning ? In human embryonic stages, which organs are formed during third to sixth week of gestation ? **1 + 1 = 2**

23. Explain co-dominance with example. **2**

24. Write the pathogen of Tetanus. Mention one symptom and a preventive measure of it. **1+ ½ + ½ = 2**

SECTION – C

25. Describe the structure of cellular endosperm with diagram. **1+2=3**

26. Define crossing over. Explain crossing over with diagram. **1+1+1=3**

27. Explain the osmosis and draw a labelled diagram of demonstration of osmosis by thistle funnel. **2 + 1 = 3**

OR

Explain lock and key theory of enzyme action and draw a labelled diagram of model.

SECTION – D

28. (i) In which plants Hatch-Slack cycle takes place ?

(ii) Explain mechanism of C_4 cycle.

(iii) Draw a line diagram of C_4 cycle.

1 + 2 + 1 = 4

OR

(i) What is nitrification ?

(ii) Explain nitrification and denitrification.

(iii) Draw a line diagram of nitrogen cycle.

29. (i) Draw a labelled diagram of internal structure of human ear.

(ii) Explain the mechanism of hearing.

(iii) Write any two diseases/defects of ear.

OR

(i) Define reflex action.

(ii) Explain mechanism of reflex action.

(iii) Draw a line diagram of reflex action.

1+2+1=4

30. (i) Draw a labelled diagram of mechanism of inspiration and

expiration.

(ii) Explain mechanism of inspiration and expiration.

OR

(i) Draw a labelled diagram of internal structure of human heart.

(ii) Describe any three diseases related to heart and circulatory' system.

1 + 3 = 4